KS3 DIGITAL TECHNOLOGY

QUANTUM COMPUTING

A new kind of computer that uses the strange rules of the very small: qubits, superposition, entanglement and interference, how quantum computers are built, what they could do, and what they can't.

LESSONS

1 From bits to qubits
2 Superposition and measurement
3 Entanglement and interference
4 Building a quantum computer
5 What could they do?
6 The quantum future

LESSON 1 OF 6

FROM BITS TO QUBITS

Ordinary computers store information in bits that are always either 0 or 1.

STARTER · 5 MINUTES

DO NOW

A coin on the table shows heads or tails. What about a coin spinning in the air? Is it heads or tails?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain the difference between a classical bit and a qubit

SUCCESS CRITERIA: I CAN...

- ✓ I can explain what a bit is
- ✓ I can explain what a qubit is using the spinning-coin model
- ✓ I can explain why quantum computers are not just "faster computers"

BITS: DEFINITELY 0 OR 1

- Every ordinary (classical) computer uses bits
- A bit is always either 0 or 1, like a light switch: off or on
- Everything (photos, music, games) is made of billions of bits
- Your phone's processor flips bits billions of times a second

KEY WORDS

BIT	The smallest unit of data in an ordinary computer: always either 0 or 1.
QUBIT	A quantum bit, which can be 0, 1 or a superposition of both until it is measured.
QUANTUM COMPUTER	A computer that uses qubits and the rules of quantum physics to solve certain problems.
CLASSICAL COMPUTER	An ordinary computer that uses bits which are always either 0 or 1.

QUBITS: 0 AND 1 AT ONCE

- A qubit is a quantum bit
- It can be in a superposition: a mixture of 0 and 1
- Like a spinning coin: not heads or tails until it lands
- When you measure it, it "lands" on 0 or 1

ACTIVITY · 15 MINUTES · PAIRS

COINS AND QUBITS

1. Place a coin heads-up: this is a bit storing 1. Turn it over: it stores 0.
2. Now spin the coin: this models a qubit in superposition.
3. "Measure" it by slapping it flat. Record the result. Repeat 20 times.
4. Count heads and tails. What do you notice?

In your workbook, answer:

? How many heads and tails did you get out of 20 spins?

? Explain one way a spinning coin is like a qubit and one way the model isn't perfect.

Your workbook: Lesson 1 · tick each step off as you go

A DIFFERENT KIND OF COMPUTER

- Not just a faster laptop: a completely different way of computing
- Uses the rules of quantum physics: the physics of atoms, electrons and light
- Brilliant at some special problems, no better at others
- You won't play games or write essays on one

QUICK CHECK

1. What values can a classical bit have?

A Only 0 or 1
B 0, 1 or 2
C Any number
D Only 1

2. What is a qubit?

A A broken bit
B A quantum bit that can be in a superposition of 0 and 1
C Eight bits
D A very small bit

3. In the spinning-coin model, what is "measuring" the qubit?

A Painting the coin
B Weighing the coin
C Spinning it faster
D Slapping the coin down to see heads or tails

4. Which statement about quantum computers is true?

A They are just laptops with more memory
B They don't exist
C They're brilliant at some special problems, but not faster at everything
D They're faster at every task

QUICK CHECK: ANSWERS

1. **A: Only 0 or 1**
   Bits are binary.

2. **B: A quantum bit that can be in a superposition of 0 and 1**
   Qubits can be in superposition.

3. **D: Slapping the coin down to see heads or tails**
   Measurement gives a definite answer.

4. **C: They're brilliant at some special problems, but not faster at everything**
   They're specialised machines.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain the difference between a bit and a qubit.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain what a bit is
- ✓ I can explain what a qubit is using the spinning-coin model
- ✓ I can explain why quantum computers are not just "faster computers"

LESSON 2 OF 6

SUPERPOSITION AND MEASUREMENT

A qubit in superposition has a certain probability of being measured as 0 or 1.

STARTER · 5 MINUTES

DO NOW

Two light switches can be in four combinations (off-off, off-on, on-off, on-on). How many combinations for three switches? For ten?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe superposition and measurement using probabilities

SUCCESS CRITERIA: I CAN...

- ✓ I can explain superposition using probabilities
- ✓ I can explain what happens when a qubit is measured
- ✓ I can calculate how many states n qubits can represent

NOT 50:50 EVERY TIME

- A qubit's superposition can be weighted: e.g. 80% chance of 0, 20% chance of 1
- Each measurement gives just one answer
- Repeat many times to see the probabilities
- Quantum programs run many times and look at the pattern of results

KEY WORDS

SUPERPOSITION	When a qubit is in a combination of 0 and 1 at the same time.
MEASUREMENT	Reading a qubit's value, which gives 0 or 1 and ends its superposition.
PROBABILITY	How likely something is to happen, often given as a percentage.

LOOKING CHANGES IT

- Measuring a qubit forces it into 0 or 1
- The superposition is lost: measuring again gives the same answer
- So you can't just "peek" during a calculation
- Even accidental "measurement" by the environment ruins calculations (lesson 4)

DOUBLING POWER

- 1 qubit: a superposition of 2 states
- 2 qubits: 4 states; 3 qubits: 8 states
- n qubits: 2^n states
- 300 qubits: more states than there are atoms in the observable universe

ACTIVITY · 20 MINUTES · GROUPS OF 3

WEIGHTED COINS

1. Use the spinner your teacher gives you: three-quarters is coloured "0" and one quarter "1".
2. Spin it 40 times and record each result.
3. Calculate the percentage of 0s and 1s and compare with the expected 75% and 25%.
4. Combine results with the whole class. Do they get closer to 75:25?

In your workbook, answer:

? Write your group's results and percentages.

? Why did combining the class results get closer to the expected probabilities?

Your workbook: Lesson 2 · tick each step off as you go

QUICK CHECK

1. What happens to a qubit's superposition when you measure it?

A It doubles
B Nothing
C It's lost: the qubit becomes 0 or 1
D It turns into two qubits

2. How many states can 4 qubits represent in superposition?

A 16
B 32
C 4
D 8

3. A qubit has a 90% chance of 0. Measured 100 times, about how many 1s?

A 90
B 10
C 0
D 50

4. Why are quantum programs run many times?

A To warm up the computer
B Each measurement gives just one random result, so you need the pattern
C To use more electricity
D Because they crash

QUICK CHECK: ANSWERS

1. C: It's lost: the qubit becomes 0 or 1

Measurement collapses it.

2. A: 16

$2^4 = 16$.

3. B: 10

10% of 100 = 10.

4. B: Each measurement gives just one random result, so you need the pattern

Repeated shots reveal probabilities.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain what happens when a qubit in superposition is measured.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain superposition using probabilities
- ✓ I can explain what happens when a qubit is measured
- ✓ I can calculate how many states n qubits can represent

LESSON 3 OF 6

ENTANGLEMENT AND INTERFERENCE

Entangled qubits are linked so that measuring one tells you about the other, however far apart they are.

STARTER · 5 MINUTES

DO NOW

Two magic dice always land on the same number, even in different rooms. How might you use dice like that?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe entanglement and interference using simple models

SUCCESS CRITERIA: I CAN...

- ✓ I can describe entanglement using the linked-dice model
- ✓ I can describe interference using waves
- ✓ I can explain how interference helps quantum algorithms

ENTANGLEMENT

- Two qubits can be entangled: their results are linked
- Measure one and you instantly know the other's result
- Works even if they're far apart
- Einstein called it "spooky action at a distance"

KEY WORDS

ENTANGLEMENT	A link between qubits so that their measurement results are connected, even when far apart.
INTERFERENCE	When waves (or quantum states) add up or cancel out.
QUANTUM ALGORITHM	A step-by-step set of operations on qubits designed to solve a problem.

INTERFERENCE

- Waves can add up (crest + crest = bigger wave)
- Or cancel out (crest + trough = flat water)
- Quantum states behave like waves and can interfere
- Noise-cancelling headphones use the same idea with sound

STEERING TOWARDS THE ANSWER

- A quantum algorithm is a recipe of operations on qubits
- It's designed so wrong answers cancel out
- And right answers add up
- So when you measure, you're very likely to get the right answer

ACTIVITY · 20 MINUTES · PAIRS

WAVE MATHS

1. Your teacher gives you strips showing waves as numbers (+2 for a crest, -2 for a trough, 0 for flat).

2. Add pairs of waves together, position by position. Which pairs cancel and which add up?

3. Play the "quantum maze": paths to wrong exits have +1 and -1 values that cancel; paths to the right exit add up.

4. Explain which exit the quantum algorithm would find.

In your workbook, answer:

? What happened when you added a wave of +2, -2, +2, -2 to one of -2, +2, -2, +2?

? Explain in your own words how interference helps a quantum computer find the right answer.

QUICK CHECK

1. What is entanglement?

A Two computers on a network

B A link between qubits so their measurement results are connected

C A type of error

D Qubits getting tangled in wires

2. What did Einstein call entanglement?

A The theory of everything

B Quantum magic

C Spooky action at a distance

D Relativity

3. What happens when a wave crest meets a wave trough of the same size?

A They cancel out

B They make a bigger crest

C They become a qubit

D Nothing

4. How do quantum algorithms use interference?

A They delete qubits

B They avoid measurement

C They make computers louder

D Wrong answers cancel and right answers add up

QUICK CHECK: ANSWERS

1. **B: A link between qubits so their measurement results are connected**
   Entangled results are linked.

2. **C: Spooky action at a distance**
   He found it very strange.

3. **A: They cancel out**
   Destructive interference.

4. **D: Wrong answers cancel and right answers add up**
   Interference steers to the answer.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe what is meant by entanglement.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe entanglement using the linked-dice model
- ✓ I can describe interference using waves
- ✓ I can explain how interference helps quantum algorithms

LESSON 4 OF 6

BUILDING A QUANTUM COMPUTER

Qubits can be made from superconducting circuits, trapped ions, atoms or photons.

STARTER · 5 MINUTES

DO NOW

Why might a computer need to be colder than outer space?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe how quantum computers are built and why errors and decoherence are major challenges

SUCCESS CRITERIA: I CAN...

- ✓ I can describe two ways of building qubits
- ✓ I can explain decoherence and why it causes errors
- ✓ I can explain what quantum error correction is

WAYS TO MAKE QUBITS

- Superconducting circuits: tiny loops of metal cooled close to absolute zero (Google, IBM)
- Trapped ions: charged atoms held by electric fields and controlled with lasers (UK and US companies)
- Neutral atoms and photons (particles of light) are other approaches
- No one knows yet which approach will win

KEY WORDS

SUPERCONDUCTING QUBIT	A qubit made from a tiny electrical circuit with no resistance, cooled close to absolute zero.
DECOHERENCE	When qubits lose their quantum state because they are disturbed, causing errors.
QUANTUM ERROR CORRECTION	Combining many physical qubits to make a more reliable logical qubit that detects and fixes errors.
ABSOLUTE ZERO	The coldest possible temperature: -273.15 °C.

COLDER THAN SPACE

- Superconducting qubits work at about -273°C, just above absolute zero
- That's colder than outer space
- Huge "chandelier" fridges called dilution refrigerators keep them cold
- Any heat or vibration can disturb the qubits

THE BIG PROBLEM: ERRORS

- Decoherence: qubits lose their superposition when disturbed
- Today's qubits make errors far more often than ordinary bits
- Quantum error correction combines many physical qubits into one reliable "logical" qubit
- In 2024 Google showed errors falling as more qubits were added, a key milestone

ACTIVITY · 25 MINUTES · PAIRS

QUANTUM COMPUTER FACT FILE

1. Choose one qubit technology from the table.
2. Use the fact sheets your teacher provides (or approved websites) to research it.
3. Make a one-page fact file: how it works, one advantage, one challenge, one organisation working on it.
4. Compare with a pair who chose a different technology.

In your workbook, answer:

? Write your fact file's advantage and challenge.

? Explain why decoherence is one of the biggest challenges in building quantum computers.

Your workbook: Lesson 4 · tick each step off as you go

QUICK CHECK

1. Why are superconducting quantum computers kept so cold?

A To save electricity
B Heat disturbs the fragile qubits
C To stop them catching fire
D Because they're made of ice

2. What is absolute zero?

A 0 °C
B The freezing point of water
C The temperature of space
D The coldest possible temperature, -273.15 °C

3. What is decoherence?

A Entangling two qubits
B A type of qubit
C Qubits losing their quantum state because they are disturbed
D Cooling a computer

4. What does quantum error correction do?

A Deletes errors from reports
B Removes the need for cooling
C Combines many physical qubits into a more reliable logical qubit
D Makes qubits bigger

QUICK CHECK: ANSWERS

1. **B: Heat disturbs the fragile qubits**
Cold reduces decoherence.

2. **D: The coldest possible temperature, -273.15 °C**
Nothing can be colder.

3. **C: Qubits losing their quantum state because they are disturbed**
Disturbance causes errors.

4. **C: Combines many physical qubits into a more reliable logical qubit**
It fights errors.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe two ways that qubits can be made.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe two ways of building qubits
- ✓ I can explain decoherence and why it causes errors
- ✓ I can explain what quantum error correction is

LESSON 5 OF 6

WHAT COULD THEY DO?

Quantum computers could simulate molecules for new medicines and batteries, help with some optimisation problems, and break today's public-key encryption.

STARTER · 5 MINUTES

DO NOW

Which would you rather have a quantum computer for: designing a new medicine, or editing your holiday photos?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Evaluate which problems quantum computers could help with and which they cannot

SUCCESS CRITERIA: I CAN...

- ✓ I can describe three possible uses of quantum computers
- ✓ I can explain the threat to encryption
- ✓ I can spot exaggerated claims about quantum computing

NATURE IS QUANTUM

- Molecules follow quantum rules, so ordinary computers struggle to simulate them
- Quantum computers could model molecules exactly
- New medicines, better batteries, greener fertilisers
- This was Richard Feynman's original idea for quantum computers (1981)

KEY WORDS

SIMULATION	Using a computer to model how something in the real world behaves.
OPTIMISATION	Finding the best solution among a very large number of possibilities.
QUANTUM ADVANTAGE	When a quantum computer solves a problem faster than any ordinary computer could.

OTHER POSSIBLE USES

- Optimisation: finding the best option among huge numbers of choices (routes, schedules)
- Some types of machine learning
- Breaking RSA encryption (Shor's algorithm)
- Searching unsorted data somewhat faster (Grover's algorithm)

HYPE CHECK

- Quantum computers won't make games, videos or web browsing faster
- Most useful applications need bigger, more reliable machines than exist today
- "Quantum advantage" claims are often for specially chosen problems
- Ask: what problem, compared with what, and has it been checked?

ACTIVITY · 20 MINUTES · GROUPS OF 3

REAL OR HYPE?

1. Read the 10 headline cards about quantum computing (some real, some exaggerated, some nonsense).
2. Sort them into Real, Promising but not yet, and Hype.
3. For two "hype" headlines, rewrite them to be accurate.

In your workbook, answer:

? Rewrite one hype headline accurately.

? Explain why simulating molecules is a good job for a quantum computer.

Your workbook: Lesson 5 · tick each step off as you go

QUICK CHECK

1. Why could quantum computers help design new medicines?

A They can test medicines on people
B They make chemicals in the lab
C They print pills
D Molecules follow quantum rules, which quantum computers can simulate

2. What is optimisation?

A Finding the best option among huge numbers of possibilities
B Making a computer look nicer
C Deleting files
D Charging a battery

3. Which algorithm could break RSA on a big quantum computer?

A Google search
B Caesar cipher
C Bubble sort
D Shor's algorithm

4. Which claim is hype?

A Quantum computers will make all your games faster
B Quantum computers could help simulate molecules
C Quantum computers threaten RSA
D Quantum computers are hard to build

QUICK CHECK: ANSWERS

1. D: Molecules follow quantum rules, which quantum computers can simulate

Simulation is a natural fit.

2. A: Finding the best option among huge numbers of possibilities

E.g. best delivery routes.

3. D: Shor's algorithm

Shor's algorithm factors numbers.

4. A: Quantum computers will make all your games faster

They don't speed up everyday tasks.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe three possible future uses of quantum computers.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe three possible uses of quantum computers
- ✓ I can explain the threat to encryption
- ✓ I can spot exaggerated claims about quantum computing

LESSON 6 OF 6

THE QUANTUM FUTURE

Quantum technology is more than computers: quantum sensors, clocks and secure communication are already being developed.

STARTER · 5 MINUTES

DO NOW

If GPS stopped working tomorrow, what would go wrong? How else could a ship or plane know where it is?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe how quantum technologies could change society and the careers they create

SUCCESS CRITERIA: I CAN...

- ✓ I can describe quantum sensors and quantum communication
- ✓ I can describe the UK's plans for quantum technology
- ✓ I can identify careers in quantum technology

BEYOND COMPUTERS

- Quantum sensors measure tiny changes in gravity, magnetism and time
- Quantum navigation could find your position without GPS
- Quantum brain scanners could be wearable
- Quantum key distribution sends encryption keys that reveal any eavesdropper

KEY WORDS

QUANTUM SENSOR	A device that uses quantum effects to measure tiny changes in things like gravity, magnetism or time.
QUANTUM KEY DISTRIBUTION	Sending encryption keys using single particles of light so that any eavesdropping can be detected.
NATIONAL QUANTUM STRATEGY	The UK government's 10-year plan (2023) to make the UK a leading quantum technology economy.

THE UK'S QUANTUM PLAN

- UK National Quantum Strategy (2023): a 10-year plan to be a leading quantum economy
- Billions of pounds of investment in research, companies and skills
- National Quantum Computing Centre in Oxfordshire
- Research hubs across UK universities, including Queen's University Belfast

QUANTUM CAREERS

- Physicists, engineers, programmers, mathematicians, chemists
- Not just scientists: business, law, ethics and teaching too
- Programming quantum computers uses languages like Python with libraries such as Qiskit
- Subjects to keep: maths, physics, computing

ACTIVITY · 25 MINUTES · GROUPS OF 3

QUANTUM 2045

1. Imagine it's 2045 and large, reliable quantum computers and sensors exist.
2. Choose one area: health, transport, the environment or security.
3. Write a short news report describing one change quantum technology has made, one benefit and one problem it caused.
4. Share and vote for the most realistic report.

In your workbook, answer:

? Write your headline and the key benefit and problem.

? Which job in quantum technology most interests you, and which school subjects would help?

Your workbook: Lesson 6 · tick each step off as you go

QUICK CHECK

1. What can a quantum sensor do?

A Make phone calls
B Store photos
C Cool a computer
D Measure tiny changes in gravity, magnetism or time

2. What is the UK National Quantum Strategy?

A A school timetable
B A 10-year plan to make the UK a leading quantum economy
C A type of qubit
D A computer game

3. How could quantum navigation help if GPS fails?

A It uses Wi-Fi
B It uses paper maps
C It senses movement and gravity to work out position without satellites
D It can't help

4. Which school subjects are most useful for quantum careers?

A Maths, physics and computing
B Art only
C PE only
D None

QUICK CHECK: ANSWERS

1. **D: Measure tiny changes in gravity, magnetism or time**
Sensors measure tiny changes.

2. **B: A 10-year plan to make the UK a leading quantum economy**
It's a government plan.

3. **C: It senses movement and gravity to work out position without satellites**
It doesn't need satellites.

4. **A: Maths, physics and computing**
These are the core subjects.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe one use of quantum technology other than quantum computers.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe quantum sensors and quantum communication
- ✓ I can describe the UK's plans for quantum technology
- ✓ I can identify careers in quantum technology

END OF TOPIC

QUANTUM COMPUTING

Revise each lesson in your workbook: the notes, key words and quick checks.
Then take the topic test when Mr Butler sets it.